Branching is not a coproduct of branches:
unfolding versus descent in the container bifibration

MacBeth
Kodamai — internal note (for Rick and Neil Ghani); CC Robin Langer

10 October 2026 (corrected revision)
corrects content commit `8972231`; content commit `4e7264d`

Abstract

The container bifibration $\mathsf{Cont}(\mathrm{cod}) = \mathrm{Fam}(\mathrm{cod}^{\mathrm{op}})$ is a candidate categorical home for the arboreal / game-comonad analysis of structure and power. In an earlier note we matched, over the base of *paths*, the arboreal relational fibration to a *one-shape* container functor, up to a fibrewise opposite. The obvious way to extend the match from paths to branching trees is to make the branches of a tree into the *shapes* of a container. We show this is wrong, and precisely why: the branch assignment $\Psi(T) = (\mathrm{br}(T), \{\mathrm{chain}_b\})$ computes the colourings of the *branch-unfolding* of the tree — the disjoint union of its root-to-leaf paths — and so over-counts the colourings of the tree itself by the exact factor $2^{\sum_v (\lambda(v)-1)}$, where $\lambda(v)$ is the number of branches through the node v. The arboreal fibre reappears inside the container fibre as the sub-poset of *prefix-consistent* colourings: the object of descent data for the cover of the tree by its branches, an equalizer that the shape-coproduct $\mathrm{Fam} = \Sigma$ cannot impose. The faithful encoding uses instead a *single* shape whose position set is the entire node set, with the tree order carried by the *base*; the encoding functor is *contravariant*, $\Psi' \colon \mathcal{T}^{\mathrm{op}} \to \mathsf{Cont}$, sending a subtree embedding to its node injection read *backward*, and the match is an isomorphism of indexed posets $T \mapsto (2^{\mathrm{nodes}(T)}, \subseteq)$. The one fact this needs — that the arboreal cartesian lift along a branching subtree embedding is computed by preimage — is proved in six lines inside our own category, with no arboreal axioms and no appeal to Jakl–Reggio (the six-line proof is Rick's, from his referee report). Branches do not transform functorially at all: the object assignment Ψ extends to no functor covering the embeddings (a no-go, either variance); the relation between the branches of T and of T' is a *span*, which is exactly why gluing is descent and not a coproduct. The reusable principle: shared sub-trajectories in a branching process are descent data, not a coproduct over branches.

1 Introduction

The bridge, and the natural guess

Two compositional vocabularies have grown up side by side. On one side, *containers* and *directed containers* — equivalently, small categories [1] — package a system as a set of *shapes* together with, for each shape, a set of *positions*; they organise into the bifibration $\mathsf{Cont}(\mathrm{cod}) = \mathrm{Fam}(\mathrm{cod}^{\mathrm{op}})$ whose fibres are predicates on positions and whose two reindexing legs are direct and inverse image [5]. On the other side, *game comonads* and the *arboreal* axiomatics [3] present model-comparison games (pebble, Ehrenfeucht–Fraïssé, modal) as comonads whose coalgebras are trees of plays, with a forgetful fibration $\mathbb{P} \colon \mathcal{A} \to \mathcal{T}$ from coloured trees to trees.

That these two pictures should be *the same picture* is a natural conjecture: both are theories of "a process together with a predicate on its states, pulled back along sub-trajectories". In an earlier

note (Stage 1, [7]) we tested it at the smallest honest scale — a single unary colour, propositionally truncated, over the base $\mathcal{P}$ of *paths* (chains of plays) — and found a clean match, up to a twist. The arboreal path fibration $\mathbb{P}|_{\mathcal{P}}$ is isomorphic to the *fibrewise opposite* of a one-shape container functor, with arboreal relation-reflection matching the *opcartesian* (pullback) leg ρ^* of $\mathsf{Cont}(\mathrm{cod})$ — not, as one might first guess, the cartesian leg. (We will see that the honest way to say this is that the encoding functor is *contravariant*; §2.)

Stage 1 reached only paths, because a one-shape container $(1, \{[n]\})$ *is* a single chain of positions and a branching tree is not. The registered next step — Stage 2 — conjectured the obvious repair: to reach the full tree base $\mathcal{T}$, make the branches of a tree into the shapes of a container,

$$\Psi(T) = \big(\mathrm{br}(T), \{\mathrm{chain}_b\}_{b \in \mathrm{br}(T)}\big),$$

one shape per root-to-leaf path, its positions the nodes along that path. The slogan was "the shape-coproduct Fam = Σ recovers the branch bookkeeping".

The result

The slogan is backwards, and the way in which it is backwards is the content of this note.

Theorem 1 (Stage 2, the three-way dividing line)**.** *In the unary, propositionally truncated regime, over the base $\mathcal{T}$ of finite rooted trees and rooted-subtree embeddings:*

(1) (Positive, one shape — proved, unconditional.) *The arboreal relational fibration extends its Stage 1 match to* all *of $\mathcal{T}$, branching included, via the* contravariant *one-shape functor $\Psi' \colon \mathcal{T}^{\mathrm{op}} \to \mathsf{Cont}$, $\Psi'(T) = (1, \{\mathrm{nodes}(T)\})$, $\Psi'(f) = (\mathrm{id}_1, j)$ with j the node injection read backward: there is an isomorphism of indexed posets $T \mapsto (2^{\mathrm{nodes}(T)}, \subseteq)$, $f \mapsto j^{-1}$. The one ingredient — that the arboreal cartesian lift along a branching subtree embedding is preimage — is proved directly, in six lines, inside the coloured category itself (§3).*

(2) (Negative, branches-as-shapes — proved.) *The conjectured branch assignment Ψ does* not *give the match, at the level of* objects*: its container fibre is the Boolean lattice of colourings of the branch-unfolding $\widehat{U}(T) = \bigsqcup_b \mathrm{chain}_b$, strictly larger than the arboreal fibre by the factor $2^{\sum_v (\lambda(v)-1)}$; the arboreal fibre reappears as the proper sub-poset of* prefix-consistent *colourings. Moreover Ψ extends to* no functor *covering the embeddings in either variance (a no-go, §5): branches do not transform functorially, and the branch relation is a* span*, not a map — which is exactly why recovering the tree is a descent step and not a coproduct.*

(3) (Arity — the unary one-shape functor is blind to k-ary relations.) *The specific unary functor $\Psi'(T) = (1, \{\mathrm{nodes}(T)\})$ cannot see a k-ary relation for $k \geq 2$, and collapses distinct k-ary structures (on two nodes, all* 16 *binary relations onto the same unary datum, of which there are* 4*). This is true and* trivial*, and it is* not *a ceiling of the bridge: the k-th power container $\Psi'_k(T) = (1, \{\mathrm{nodes}(T)^k\})$ reproduces the k-ary fibre exactly (§6).*

The three clauses answer three different questions, and conflating them was the error in the original conjecture. Clause (2) is about the *wrong* encoding of branching; clause (1) about the *right* one; clause (3) about an artefact of one particular functor, not an obstruction of the bridge.

The mechanism: unfolding versus descent

The heart is clause (2), and it has a one-line diagnosis. The shape-coproduct Fam = Σ is a *coproduct*: applied to the branches of a tree it forms their disjoint union $\widehat{U}(T)$, the universal

unfolding of the tree by its paths, in which every shared prefix is duplicated once per branch that passes through it. Colouring $\widehat{U}(T)$ freely lets the duplicated copies of a node disagree. The colourings of the *tree* are exactly those that agree across copies — the *prefix-consistent* ones — and reconstructing the tree from its branch-cover by imposing that agreement is a *gluing*: a descent datum, the equalizer of the coproduct, of the wrong variance for a coproduct to supply. Thus

$$\mathbb{P}\text{-fibre over } T \;=\; \text{descent data for the branch-cover } \widehat{U}(T) \twoheadrightarrow T, \qquad \mathsf{Cont}(\mathrm{cod})\circ\Psi\text{-fibre} \;=\; \text{the raw cover colouring}$$

they coincide iff the cover is trivial, i.e. T is a path — which is exactly why Stage 1 saw no discrepancy. The honest encoding, clause (1), sidesteps the unfolding altogether: keep *one* shape and put the whole node set in the positions, letting the *base* $\mathcal{T}$ carry the tree order through its subtree-embedding morphisms. One copy of each node, no duplication, nothing to glue.

Why it matters

The reusable statement is a modelling warning with teeth. Whenever a branching process — a supply chain forking into parallel routes, an agent-orchestration log splitting into concurrent sub-dialogues, a provenance DAG — is to be encoded in the container bifibration, the temptation is to let each branch be a shape and take their coproduct. Our count shows precisely what that costs: an over-representation by $2^{\sum_v(\lambda(v)-1)}$, one spurious bit of freedom for every extra branch through every shared node. The shared history that distinguishes a tree from its unfolding is *descent data*, and the Fam $=\Sigma$ coproduct structurally cannot see it. This is the container-side form of our running theme that *composition is a nerve-level-two datum*: the sharing lives one level up from the branches themselves, and is recovered by an explicit gluing step, never read off a product over branches.

Outline

Section 2 fixes the two fibrations and recalls the Stage 1 orientation in its contravariant form. Section 3 gives the positive one-shape extension (clause 1) and the six-line proof of its one ingredient (Rick's; §3), inside the coloured category, with no appeal to Jakl–Reggio. Section 4 proves clause (2)'s object-level over-count, with the V- and Y-tree witnesses and the descent diagnosis. Section 5 proves the no-go: Ψ extends to no functor covering the embeddings, nor any natural on automorphisms. Section 6 treats clause (3) and corrects the former "ceiling". Section 7 records scope, open problems, and placement.

Note on this revision. This note corrects commit `8972231` following Rick's referee report of 10 October 2026. Three things were wrong and are now fixed, all traceable to a single mistake — the encoding functor was made covariant and then patched by hand with a nearest-ancestor retraction ρ_f, which does not compute preimage. We make Ψ' *contravariant* and carry the node injection j backward (clause 1 becomes proved and unconditional); we replace the Jakl–Reggio dependence for the cartesian-lift fact with a direct six-line proof *supplied by Rick's referee report* (that paper is not needed, and is correctly cited in §7); we state the no-go for Ψ on morphisms explicitly; and we withdraw the $k \geq 2$ "ceiling", which is false.

A second round of edits (11 October 2026) follows Rick's round-2 report. **M-A:** Lemma 8, its six-line proof, and the exhaustive ≤ 4-node (3147-morphism) check are explicitly credited to Rick throughout (§3). **M-B:** the no-go is stated only with its hypothesis — Theorem 18 quantifies over functors *covering* the embeddings; the unrestricted "no functor at all" reading is false (constant-branch functors exist, Remark 20), so every restatement now carries the hypothesis, and we add

Rick's variance-symmetric sharpening Theorem 19 (naturality on automorphisms suffices, with no condition on the embeddings).

2 Two fibrations and the Stage 1 match

Throughout we work in the **unary, propositionally truncated** regime: a single relation symbol, interpreted as a *subset* (a subobject, not a proof-relevant family). This is the fragment in which the two theories have a chance of agreeing; §6 isolates what the unary one-shape functor cannot reach.

2.1 The container bifibration

Recall [1, 2] that a *container* is a pair $(S, \{P_s\}_{s\in S})$ of a set S of shapes and a family of position sets, and that containers organise into $\mathsf{Cont}(\mathrm{cod}) = \mathrm{Fam}(\mathrm{cod}^{\mathrm{op}})$, the bifibration over $\mathsf{Cont}(\mathsf{Set})$ whose fibre over $(S, \{P_s\})$ is $\prod_{s\in S}(\mathsf{Set}/P_s)^{\mathrm{op}}$. We use only its propositional truncation, recorded in [5]. A container morphism $(S, \{P_s\}) \to (S', \{P'_{s'}\})$ is a pair $(\varphi\colon S \to S',\ \{\rho_s\colon P'_{\varphi s} \to P_s\})$: a *forward* shape map and, for each shape, a *backward* position map. Then:

- **Fibre.** Over $(S, \{P_s\})$ the truncated fibre is $\prod_{s\in S} \mathrm{Sub}(P_s)^{\mathrm{op}} = \big(2^{\coprod_s P_s}, \supseteq\big)$: predicates on the total position set, ordered by *reverse* inclusion.
- **Cartesian leg.** The backward position map ρ has cartesian reindexing $(\Sigma_\rho)^{\mathrm{op}}$, truncated *direct image* $\rho(-)\colon \mathrm{Sub} \to \mathrm{Sub}$.
- **Opcartesian (pullback) leg.** The same ρ has opcartesian reindexing ρ^*, truncated *inverse image* ρ^{-1}, left adjoint to the cartesian leg.

The two legs are the whole point: "pull a predicate back along a sub-trajectory" is ρ^*, inverse image, *not* the cartesian direction. Note that it is the *opcartesian* leg that performs inverse image; this will force the encoding functor below to be contravariant.

2.2 The tree base and the arboreal relational fibration

Definition 2. $\mathcal{T}$ is the category of finite rooted trees of bounded height (a tree is a finite poset with least element root in which every down-set is a chain) and *rooted-subtree embeddings*: a morphism $f\colon T \to T'$ is an injection on nodes $j = \mathrm{nodes}(f)\colon \mathrm{nodes}(T) \hookrightarrow \mathrm{nodes}(T')$ whose image is prefix-closed and contains the root. A morphism of $\mathcal{T}$ is determined by its node map. $\mathcal{P} \subset \mathcal{T}$ is the full subcategory of chains (single branches); on $\mathcal{P}$ the only morphisms are initial-segment inclusions — exactly the Stage 1 base.

Definition 3. $\mathcal{A}$ is the category of trees carrying one unary colour $R \subseteq \mathrm{nodes}(T)$ — the unary, truncated relational lift of the arboreal coalgebra category, whose underlying coalgebras are the copresheaves of [6, Lemma 4.1]. A morphism $(T, R) \to (T', R')$ is a subtree embedding $f\colon T \hookrightarrow T'$ that is *colour-preserving*, $R \subseteq j^{-1}R'$ (equivalently $j(R) \subseteq R'$), and there is at most one over each base map, so $\mathcal{A} \to \mathcal{T}$ is a poset fibration. The functor $\mathbb{P}\colon \mathcal{A} \to \mathcal{T}$ forgets the colour; it is faithful.

Two facts about $\mathbb{P}$ are all we use; both are elementary.

Lemma 4 (arboreal fibre and reindexing)**.** *1. A vertical map $R \to R'$ over id_T exists iff $R \subseteq R'$. Hence the $\mathbb{P}$-fibre over T is the* full *Boolean lattice $(2^{\mathrm{nodes}(T)}, \subseteq)$ — every subset is a legal colouring, branching imposing no constraint.*

2. *Along a subtree embedding $f\colon T \hookrightarrow T'$ with node injection j, the arboreal reindexing restricts the colour: $f^*R' = R' \cap \mathrm{nodes}(T) = j^{-1}(R')$ (relation-reflection). Inverse image is strictly functorial.*

The identification in part (1) depends on reading a node of T as the unique play from the root to it: the shared root is a *single* play, so $\mathbb{P}$ colours $\mathrm{nodes}(T)$ with the prefix shared, not the branch-disjoint unfolding. This is the sentence the branch assignment Ψ will violate.

2.3 The Stage 1 orientation, contravariantly

Lemma 5 (Stage 1, [7])**.** *Present the Stage 1 encoding as a* contravariant *functor $\Phi\colon \mathcal{P}^{\mathrm{op}} \to \mathsf{Cont}(\mathsf{Set})$, $\Phi([n]) = (1, \{[n]\})$, sending an initial-segment inclusion $[m] \hookrightarrow [n]$ to the container morphism $(\mathrm{id}_1, j)\colon \Phi([n]) \to \Phi([m])$ with backward position map the inclusion $j\colon [m] \hookrightarrow [n]$. Then the $\mathbb{P}$-fibre $(2^{[n]}, \subseteq)$ and the truncated container fibre $\mathrm{Sub}([n])^{\mathrm{op}} = (2^{[n]}, \supseteq)$ are* opposite *posets, and arboreal relation-reflection j^{-1} equals the* opcartesian *leg ρ^* of $\mathsf{Cont}(\mathrm{cod})$ along (id_1, j) (differing from the cartesian $(\Sigma_j)^{\mathrm{op}}$ = direct image). Hence $\mathbb{P}|_{\mathcal{P}} \cong (\Phi^*\mathsf{Cont}(\mathrm{cod}))^{\text{fib-op}}$ over $\mathcal{P}^{\mathrm{op}}$.*

We stress the variance, because it is where the earlier Stage 2 note went wrong. The backward map that Stage 1 carries is the *inclusion* j (not a collapse): reindexing is j^{-1}, preimage. The fibre-op and the ρ^* identification used only that j is an injection of sets and that inverse image is functorial — never that the positions formed a chain. This is why one of the two candidate extensions survives verbatim, and the other does not.

3 The positive extension: one shape, contravariant, positions the whole tree

Definition 6. $\Psi'\colon \mathcal{T}^{\mathrm{op}} \to \mathsf{Cont}(\mathsf{Set})$ sends T to the *one-shape* container $(1, \{\mathrm{nodes}(T)\})$: a single shape, position set the entire node set as a *flat* set, the tree order discarded by Ψ' and carried by the base $\mathcal{T}$. On a subtree embedding $f\colon T \hookrightarrow T'$ of $\mathcal{T}$, with node injection $j\colon \mathrm{nodes}(T) \hookrightarrow \mathrm{nodes}(T')$,

$$\Psi'(f) = (\mathrm{id}_1,\ j)\ :\ \Psi'(T') \longrightarrow \Psi'(T),$$

where j is the *backward position map* of the container morphism (there is one shape, so a morphism $\Psi'(T') \to \Psi'(T)$ is a single backward map $\mathrm{nodes}(T) \to \mathrm{nodes}(T')$, namely j). On a chain, $\Psi'([n]) = (1, \{[n]\}) = \Phi([n])$, so Ψ' extends Φ; and Ψ' is functorial because node injections compose functorially, $j_{g\circ f} = j_g \circ j_f$.

Remark 7 (Why the former ρ_f was wrong)**.** The earlier version made Ψ' covariant, $\Psi'(f) = (\mathrm{id}_1, \rho_f)$ with $\rho_f\colon \mathrm{nodes}(T') \twoheadrightarrow \mathrm{nodes}(T)$ the nearest-ancestor retraction. Neither reindexing leg along ρ_f equals relation-reflection j^{-1}. Concretely on the V-tree $T' = \{r, a, b\}$, $T = \{r, a\}$ with $\rho_f(b) = r$: for $R' = \{b\}$, $j^{-1}R' = \varnothing$ but the direct image $\Sigma_{\rho_f}R' = \{r\}$; for $R' = \{a, b\}$, $j^{-1}R' = \{a\}$ but $\Sigma_{\rho_f}R' = \{r, a\}$. The opcartesian leg ρ_f^{-1} even goes the wrong way (from T to T'). Carrying j backward, as in Definition 6, is the fix, and it is the same orientation Stage 1 already used.

3.1 The one ingredient, in six lines

To promote a fibrewise match to an isomorphism of indexed posets we need that the arboreal *cartesian lifts* along subtree embeddings are computed by preimage. We prove this directly in the coloured category $\mathcal{A}$, with no arboreal axioms and no external input.

Lemma 8 (the branching cartesian lift is preimage)**.** *In $\mathcal{A}$ (Definition 3), let $f\colon (T,R)\to(T',R')$ be over the node injection j, and recall $\mathbb{P}$ is faithful. Then:*

1. *If $R = j^{-1}R'$ then f is $\mathbb{P}$-cartesian.*
2. *If f is $\mathbb{P}$-cartesian then $R = j^{-1}R'$.*
3. *f is a pathwise embedding (strong on each root chain) iff $R = j^{-1}R'$.*

Consequently the cartesian lift of $f\colon T \hookrightarrow T'$ at R' is $(T, j^{-1}R')$ — preimage — for every *T, branching included.*

Proof. (1) Given $g\colon (S,Q)\to(T',R')$ over $k = j\circ h$: then $j(h(Q)) = k(Q)\subseteq R'$, so $h(Q)\subseteq j^{-1}R' = R$; hence h lifts to a morphism $(S,Q)\to(T,R)$, uniquely since $\mathbb{P}$ is faithful. (2) The map $(T, j^{-1}R')\to(T',R')$ over j is a morphism; cartesianness of f lifts id_T, giving $j^{-1}R'\subseteq R$. The reverse inclusion holds because f is a morphism ($j(R)\subseteq R'$, i.e. $R\subseteq j^{-1}R'$). (3) "Strong on each root chain c" means $R\cap c = j^{-1}R'\cap c$; the root chains cover T, so this is $R = j^{-1}R'$. □

The argument is uniform in T: there is no "non-path case", so no arboreal machinery is invoked. *Attribution.* Lemma 8 and the six-line proof above are due to Rick's referee report of 10 October 2026, as is the exhaustive $\leq$ 4-node enumeration that confirms it (all coloured trees with ≤ 4 nodes, 3147 morphisms, have cartesian $\Leftrightarrow$ preimage $\Leftrightarrow$ pathwise in every case); our `scratch/crown_stage2_repair_verify.py` reproduces the V-tree slice. Equivalently, $\mathbb{P}\cong \mathrm{nodes}^*(\mathrm{Sub})$: the predicate fibration $\mathrm{Sub}\to\mathsf{Set}$ (fibre $(2^X,\subseteq)$, reindexing = preimage) pulled back along $\mathrm{nodes}\colon \mathcal{T}\to\mathsf{Set}$, $T\mapsto \mathrm{nodes}(T)$, $f\mapsto j$.

3.2 The match, as an isomorphism of indexed posets

Proposition 9 (one-shape extension)**.** *In the unary truncated regime there is an isomorphism of indexed posets over $\mathcal{T}^{\mathrm{op}}$,*

$$\mathbb{P}\ \cong\ \big(\Psi'^{*}\mathsf{Cont}(\mathrm{cod})\big)^{\text{fib-op}}_{\text{opfib}},\qquad T\longmapsto (2^{\mathrm{nodes}(T)},\subseteq),\quad f\longmapsto j^{-1},$$

obtained from the opfibration part *of the pullback $\Psi'^{*}\mathsf{Cont}(\mathrm{cod})$ over $\mathcal{T}^{\mathrm{op}}$ after the fibrewise opposite. The branching of T is encoded entirely in the base and its subtree-embedding morphisms, while the container functor keeps one flat position set.*

Proof. Fibrewise: over $\Psi'(T) = (1,\{\mathrm{nodes}(T)\})$ the truncated container fibre is $\mathrm{Sub}(\mathrm{nodes}(T))^{\mathrm{op}} = (2^{\mathrm{nodes}(T)},\supseteq)$, whose fibrewise opposite is the $\mathbb{P}$-fibre $(2^{\mathrm{nodes}(T)},\subseteq)$ of Lemma 4(1), on the nose — one shape, one copy of each node, no over-count. On morphisms: along $\Psi'(f) = (\mathrm{id}_1, j)$ the opcartesian leg of $\mathsf{Cont}(\mathrm{cod})$ is j^{-1} = inverse image, which equals arboreal relation-reflection (Lemma 4(2)), and Lemma 8 identifies the arboreal cartesian lift with the same preimage. The identification uses only that j is a set injection and that inverse image is functorial; no step uses a chain order. Composing gives the stated isomorphism. □

Remark 10 (Well-typedness; nothing is conditional)**.** The earlier phrase "$\mathbb{P}\cong(\Psi'^{*}\mathsf{Cont}(\mathrm{cod}))_{\text{fib-op}}$ as fibrations over $\mathcal{T}$" was ill-typed, because the variance is wrong: the encoding is contravariant, so the base is $\mathcal{T}^{\mathrm{op}}$ and the correct object is an isomorphism of indexed posets $\mathcal{T}^{\mathrm{op}}\to\mathbf{Pos}$, as above. With Ψ' contravariant, the word *conditional* leaves the statement entirely: clause (1) is proved by Lemma 8 together with Proposition 9, with no appeal to any external cartesian-lift theorem.

4 Branches as shapes unfolds the tree

This section proves the object-level half of clause (2): the branch assignment computes the colourings of the wrong object. Nothing here is conditional. (That Ψ is moreover not a functor is §5.)

Definition 11. Ψ sends a tree T to the container $(\mathrm{br}(T), \{\mathrm{chain}_b\}_b)$: one shape per root-to-leaf branch b, with position set over b the chain of nodes along b. On a chain there is one branch and Ψ agrees with $\Phi = \Psi'$, so Stage 1 holds; off $\mathcal{P}$ they diverge. We treat Ψ as an *object assignment* here; §5 shows it is no functor.

Definition 12. The *branch-unfolding* of T is $\widehat{U}(T) = \bigsqcup_{b \in \mathrm{br}(T)} \mathrm{chain}_b$, the disjoint union of the branch-chains, with *gluing map* $\pi \colon \widehat{U}(T) \twoheadrightarrow \mathrm{nodes}(T)$, $(b, v) \mapsto v$. For a node v, write $\lambda(v)$ for the number of branches through v (equivalently, the number of leaves above v), so $\pi^{-1}(v)$ has size $\lambda(v)$.

Lemma 13 (the exact over-count, an objectwise identity)**.** *The truncated container fibre over $\Psi(T)$ is $\prod_b \mathrm{Sub}(\mathrm{chain}_b)^{\mathrm{op}} = (2^{\widehat{U}(T)}, \supseteq)$, of cardinality $2^{|\widehat{U}(T)|}$ with*

$$|\widehat{U}(T)| = \sum_b |\mathrm{chain}_b| = \sum_{v \in \mathrm{nodes}(T)} \lambda(v), \qquad |\widehat{U}(T)| - |\mathrm{nodes}(T)| = \sum_v (\lambda(v) - 1).$$

The pullback $\pi^* \colon 2^{\mathrm{nodes}(T)} \hookrightarrow 2^{\widehat{U}(T)}$ is injective with image the prefix-consistent *colourings — those constant on each fibre $\pi^{-1}(v)$. Hence the $\mathbb{P}$-fibre $(2^{\mathrm{nodes}(T)}, \subseteq)$ embeds as the proper sub-poset of prefix-consistent colourings inside the $\mathsf{Cont}(\mathrm{cod}) \circ \Psi$-fibre $(2^{\widehat{U}(T)}, \supseteq)$, the excess being the factor $2^{\sum_v (\lambda(v)-1)}$. The inclusion is proper exactly when some node has $\lambda(v) \geq 2$, i.e. exactly when T branches.*

Proof. The fibre formula is the container fibre with $\bigsqcup_s P_s = \bigsqcup_b \mathrm{chain}_b = \widehat{U}(T)$. Counting positions by the node they glue to gives $|\widehat{U}(T)| = \sum_v \lambda(v)$. A subset $A \subseteq \widehat{U}(T)$ equals $\pi^*(B)$ iff A is a union of fibres of π (constant on each $\pi^{-1}(v)$); such $B = \{v : \pi^{-1}(v) \subseteq A\}$ is then unique, so π^* is injective with the stated image. The complementary degrees of freedom number $\prod_v 2^{\lambda(v)-1} = 2^{\sum_v (\lambda(v)-1)}$, positive iff some $\lambda(v) \geq 2$. □

This is a one-line count; it is a comparison of fibres *over single objects*, and it holds regardless of whether Ψ acts on morphisms. In particular the two fibres have different cardinalities as soon as T branches, so there is *no* fibrewise equivalence $\mathbb{P} \cong (\Psi^* \mathsf{Cont}(\mathrm{cod}))^{\mathrm{op}}$ — not even onto the image of Ψ. (Verified on all 154 labelled rooted trees with ≤ 6 nodes; `scratch/crown_stage2_repair_verify.py`.)

Example 14 (the V-tree)**.** Let V have root r and two leaves a, b. Then $|\mathrm{nodes}| = 3$, the branches are ra, rb, $|\widehat{U}| = 4$, $\lambda(r) = 2$, $\lambda(a) = \lambda(b) = 1$, so $\sum_v (\lambda(v) - 1) = 1$. The arboreal fibre has $2^3 = 8$ colourings; the $\mathsf{Cont}(\mathrm{cod}) \circ \Psi$-fibre has $2^4 = 16$. The 8 excess colourings are the *inconsistent* ones — colouring the root in branch ra but not in rb, or vice versa.

Example 15 (the Y-tree: deeper sharing)**.** Let Y have root r, a single child c, and c in turn with two children x, y. Then $|\mathrm{nodes}| = 4$, the branches are rcx, rcy, $|\widehat{U}| = 6$, $\lambda(r) = \lambda(c) = 2$, $\lambda(x) = \lambda(y) = 1$, so $\sum_v (\lambda(v) - 1) = 2$: the whole shared spine $\{r, c\}$ is double-counted. The arboreal fibre has $2^4 = 16$ colourings, the $\mathsf{Cont}(\mathrm{cod}) \circ \Psi$-fibre $2^6 = 64$, ratio $2^2 = 4$.

4.1 The diagnosis: Fam = Σ is unfolding, gluing is descent

Lemma 13 is a count; here is what it means. The branch family $\{\mathrm{chain}_b \hookrightarrow T\}_b$ is a *cover* of T by its paths, for the *Alexandrov topology of down-sets*: each root chain is a down-set, hence open, and the chains cover T. Its overlaps are $\mathrm{chain}_b \cap \mathrm{chain}_{b'} = \mathrm{chain}_{b\wedge b'}$, the shared prefix down to the meet of b and b'. A colouring of the tree is the same as a family of colourings of the branches that *agree on overlaps*: a *descent datum* for the cover.

Proposition 16 (descent description of the arboreal fibre)**.** *For every T, the image of π^* — the prefix-consistent colourings — equals the set of colourings of $\widehat{U}(T)$ that agree on all pairwise overlaps of the branch cover; and this equals $2^{\mathrm{nodes}(T)}$, the full arboreal fibre. Thus*

$$\mathbb{P}\text{-fibre over } T \;=\; \text{descent data (the equalizer) for } \widehat{U}(T) \twoheadrightarrow T, \qquad \mathsf{Cont}(\mathrm{cod})\circ\Psi\text{-fibre} \;=\; \text{the raw cover colourings,}$$

coinciding iff the cover is trivial, i.e. T a path.

Proof. Two copies (b, v), (b', v) of a node v lie in the overlap $\mathrm{chain}_{b\wedge b'}$ exactly when both b, b' pass through v; as b, b' range over branches through v these exhaust $\pi^{-1}(v)$. Agreement on all overlaps therefore says a colouring is constant on each $\pi^{-1}(v)$ — prefix-consistency — which by Lemma 13 is the image of π^*, in bijection with $2^{\mathrm{nodes}(T)}$. The gluing is the single equalizer $2^{\mathrm{nodes}(T)} \rightarrowtail 2^{\widehat{U}(T)} \rightrightarrows \prod_{b,b'} 2^{\mathrm{chain}_{b\wedge b'}}$ (verified in `scratch/crown_stage2_repair_verify.py`: consistent = agree-on-overlap = 2^{nodes}). □

The conceptual point is a mismatch of variance. Fam = Σ is a coproduct over shapes; fed the branches it yields the disjoint union $\widehat{U}(T)$, the unfolding of the tree. Reconstructing the tree is the *opposite* operation — a gluing that identifies shared prefixes, the equalizer of the coproduct, a limit-shaped datum. A coproduct cannot impose it. So Ψ realises $\mathbb{P}$ only as the *descent sub-fibration* of prefix-consistent colourings, never as the whole $\mathsf{Cont}(\mathrm{cod}) \circ \Psi$.

Remark 17 (reconciliation with clause (1))**.** Ψ and Ψ′ differ exactly by unfolding versus gluing: $\Psi(T)$ is the branch cover, $\Psi'(T)$ is the tree. $\mathsf{Cont}(\mathrm{cod})$ colours whatever flat position set it is handed. Hand it the unfolded cover (Ψ) and it colours the cover, which descends to the tree only after an equalizer Fam cannot supply; hand it the node set (Ψ′) and it colours the tree directly. **Branching is not a coproduct of branches**: the faithful encoding is a single shape whose position set is the branching node set, with the base $\mathcal{T}$ carrying the order.

5 Branches do not transform functorially: a no-go

Lemma 13 compares fibres over single objects, and that is all the over-count needs. But Ψ was never defined on morphisms, and the question whether it *can* be is not cosmetic: it is the precise reason the branch picture is descent and not a functor. The answer is no in both variances — for any functor whose action *covers* the subtree embeddings (Theorem 18), and in fact for any functor that merely acts on automorphisms by the canonical action on branches, with no condition on the embeddings at all (Theorem 19). The compatibility hypothesis cannot, however, simply be *dropped*: with no condition relating $\Psi(f)$ to f there do exist functors $\mathcal{T} \to \mathsf{Cont}$ and $\mathcal{T}^{\mathrm{op}} \to \mathsf{Cont}$ extending Ψ on objects (Remark 20), and they are exactly the ones that act trivially on automorphisms.

Theorem 18 (no functorial action for Ψ)**.** *The object assignment $T \mapsto (\mathrm{br}(T), \{\mathrm{chain}_b\})$ extends to* no *functor $\mathcal{T} \to \mathsf{Cont}$ nor $\mathcal{T}^{\mathrm{op}} \to \mathsf{Cont}$ whose action on morphisms covers the subtree embeddings. Concretely:*

1. (Contravariant fails on existence.) *A branch of the larger tree can restrict to a non-branch prefix, leaving no valid shape image.*

2. (Covariant fails on functoriality.) *A sibling-swap automorphism forces an impossible fixed point of the branch shape-map.*

Equivalently, $\mathrm{br}\colon \mathcal{T} \to \mathsf{Set}$ *is not a functor compatible with the embeddings in either variance.*

Here "covers f" means the shape map of $\Psi(f)$ sends each branch to a branch compatible with j: for the covariant form, a branch b of the source with $j(\mathrm{chain}_b) \subseteq \mathrm{chain}_{\varphi(b)}$; for the contravariant form, a branch b' with $j(\mathrm{chain}_{\varphi(b')}) \subseteq \mathrm{chain}_{b'}$. Dropping compatibility makes $\Psi(f)$ carry no information about f, so it would not be "the action of Ψ"; we return to that at the end.

Proof. (1) On the V-tree inclusion $\{r, a\} \hookrightarrow \{r, a, b\}$: $\mathrm{br}(T') = \{ra, rb\}$, $\mathrm{br}(T) = \{ra\}$. The shape map of a contravariant $\Psi(f)\colon \Psi(T') \to \Psi(T)$ would send $b' = rb$ to a branch b of T with $j(\mathrm{chain}_b) \subseteq \mathrm{chain}_{rb} = \{r, b\}$; the only candidate $\mathrm{chain}_{ra} = \{r, a\}$ maps to $\{r, a\} \not\subseteq \{r, b\}$. The branch rb of T' restricts to the prefix $\{r\}$, which is not a branch of T. No valid image.

(2) Covariant shape maps $\varphi\colon \mathrm{br}(T) \to \mathrm{br}(T')$ with $j(\mathrm{chain}_b) \subseteq \mathrm{chain}_{\varphi(b)}$ exist per morphism (extend a branch to some leaf above), but functoriality fails. Let $T = \{r, a\}$ (a a leaf) and $T' = \{r, a, c, d\}$ with c, d both children of a; let $f\colon T \hookrightarrow T'$ be the inclusion and $\sigma\colon T' \to T'$ the automorphism swapping $c \leftrightarrow d$ (fixing r, a). Then $\mathrm{br}(T) = \{\beta_a\}$ with $\mathrm{chain}_{\beta_a} = \{r, a\}$, and $\mathrm{br}(T') = \{\beta_c = rac,\ \beta_d = rad\}$. Compatibility for f forces $\varphi_f(\beta_a) \in \{\beta_c, \beta_d\}$ — two choices. Compatibility for σ forces $\varphi_\sigma(\beta_c) = \beta_d$ and $\varphi_\sigma(\beta_d) = \beta_c$, since $\sigma(\mathrm{chain}_{\beta_c}) = \{r, a, d\} = \mathrm{chain}_{\beta_d}$: the swap, with no fixed point. But $\sigma \circ f = f$ in $\mathcal{T}$ (equal node maps, as σ fixes r, a and morphisms of $\mathcal{T}$ are determined by their node maps), so functoriality forces $\varphi_\sigma \circ \varphi_f = \varphi_f$, i.e. $\varphi_f(\beta_a)$ is a fixed point of φ_σ. The swap has none. Contradiction. □

The two halves of Theorem 18 use the covering hypothesis in different strengths: the covariant half (2) needs only that $\varphi_f(\beta_a)$ land in $\mathrm{br}(T')$ — no compatibility at all — while the contravariant half (1) uses the specific compatibility $j(\mathrm{chain}_{\varphi(b')}) \subseteq \mathrm{chain}_{b'}$ (a definitional choice, stronger than restriction $j^{-1}(\mathrm{chain}_{b'}) \subseteq \mathrm{chain}_{\varphi(b')}$). The following variance-symmetric sharpening, which we owe to Rick's referee report, drops *all* conditions on the non-invertible morphisms and asks only naturality on automorphisms.

Theorem 19 (no action natural on automorphisms; after Rick)**.** *There is no functor* $\mathcal{T} \to \mathsf{Cont}$ *nor* $\mathcal{T}^{\mathrm{op}} \to \mathsf{Cont}$ *that extends* Ψ *on objects and acts on* automorphisms *by the canonical action on branches — with no condition imposed on non-invertible morphisms.*

Proof. Covariant: the sibling swap of Theorem 18(2) applies unchanged, since $\varphi_f(\beta_a) \in \mathrm{br}(T') = \{\beta_c, \beta_d\}$ needs no compatibility — only $\varphi_\sigma =$ swap (the canonical action of σ) and $\sigma f = f$ are used, and the swap has no fixed point. Contravariant: take the three-leaf star $f\colon V = \{r; a, b\} \hookrightarrow S_3 = \{r; a, b, e\}$, with $\sigma \in \mathrm{Aut}\, V$ swapping a, b and $\tau \in \mathrm{Aut}\, S_3$ swapping a, b and fixing e, so $\tau f = f\sigma$. The canonical actions give $\varphi_\tau(re) = re$ and $\varphi_\sigma =$ swap on $\{ra, rb\}$, whence naturality forces $\varphi_f(re) \in \{ra, rb\}$ to be a fixed point of the swap; there is none. Both cases are variance-symmetric and use only automorphism naturality (checked-sober; computed, 0 functors in each variance). The constant-branch functors of Remark 20 show the hypothesis cannot be dropped entirely. □

Consequently clause (2) of Theorem 1 is honestly a statement about the *object* assignment Ψ. The genuine structure relating the branches of T and of T' is a *span* / correspondence

$$R_f \subseteq \mathrm{br}(T) \times \mathrm{br}(T'), \qquad (b, b') \in R_f \iff j(\mathrm{chain}_b) \subseteq \mathrm{chain}_{b'},$$

read backward, R_f is a left-total *injective partial map* $\mathrm{br}(T') \rightharpoonup \mathrm{br}(T)$ (if $(b,b'),(c,b') \in R_f$ then $j(\mathrm{chain}_b), j(\mathrm{chain}_c)$ are comparable maximal chains inside $\mathrm{chain}_{b'}$, hence equal, so it is left-unique), defined exactly on those b' whose restriction is a full branch of T; read forward it is multivalued (not a function). It is thus not a function in *either* direction simultaneously — which is precisely why $\mathrm{Fam} = \Sigma$ (a functor) cannot realise it, and why recovering $\mathbb{P}$ from $\mathsf{Cont}(\mathrm{cod}) \circ \Psi$ needs a *descent* step rather than a functorial map. The no-go is the container-side face of *composition is a degree-two datum*: sharing of branch-prefixes is a relation, not a map.

Remark 20 (the compatibility hypothesis is essential, and the escapes act trivially on automorphisms)**.** The hypothesis of Theorem 18 cannot be dropped *outright*: if the shape map is allowed to ignore f entirely, there *do* exist functors $\mathcal{T} \to \mathsf{Cont}$ and $\mathcal{T}^{\mathrm{op}} \to \mathsf{Cont}$ extending Ψ on objects. Passing to a skeleton of $\mathcal{T}$ (where a finite injective endomorphism is bijective, so every endomorphism is an automorphism), send each non-isomorphism to the constant map at a fixed branch $c_T \in \mathrm{br}(T)$, with position maps constant at the root, and act on automorphisms as the identity; constant maps absorb composition and identities are preserved, so this is a functor in either variance. What all such escapes have in common is that they act *trivially* on automorphisms: in (2) the only escape is $\varphi_\sigma = \mathrm{id}$, which fails to cover the nontrivial swap σ (it identifies the distinct subtrees $r \to a \to c$ and $r \to a \to d$), so it is not an encoding of $\mathcal{T}$. This is exactly the gap that Theorem 19 closes with a single hypothesis: any functor distinguishing σ from id by the canonical action on branches runs into the contradiction, in both variances and with no condition on the embeddings.

6 The arity clause, and why there is no ceiling

Everything above is $k = 1$. Clause (3) concerns $k \geq 2$, and here we must correct the earlier note, which claimed a ceiling "beyond $\mathsf{Cont}(\mathrm{cod})$". There is none.

Proposition 21 (the unary one-shape functor is blind to k-ary relations)**.** *A k-ary relation on T is a subset of* $\mathrm{nodes}(T)^k$*. For $k \geq 2$ this is not a subobject in* $\mathsf{Set}/\mathrm{nodes}(T)$*, and* $\mathsf{Cont}(\mathrm{cod})$ *over* $\Psi'(T) = (1, \{\mathrm{nodes}(T)\})$ *has fibre* $\mathrm{Sub}(\mathrm{nodes}(T))^{\mathrm{op}}$ *— unary predicates only. So this particular functor (2^n on n nodes) collapses distinct k-ary structures: on two nodes $\{0,1\}$ the relations $\varnothing$ and $\{(0,1)\}$, among all $2^4 = 16$ binary relations, map to the same datum, of which there are only $2^2 = 4$. This statement is true and* trivial *— a functor into unary predicates has no k-ary values.*

Proposition 22 (the k-th power container reproduces the k-ary fibre)**.** *Let* $\Psi'_k \colon \mathcal{T}^{\mathrm{op}} \to \mathsf{Cont}$*,* $\Psi'_k(T) = (1, \{\mathrm{nodes}(T)^k\})$*,* $\Psi'_k(f) = (\mathrm{id}_1, j^k)$ *with* $j^k = j \times \cdots \times j \colon \mathrm{nodes}(T)^k \to \mathrm{nodes}(T')^k$ *the coordinatewise backward map. Then its fibre over T is* $\mathrm{Sub}(\mathrm{nodes}(T)^k)^{\mathrm{op}} = (2^{\mathrm{nodes}(T)^k}, \supseteq)$*, and the opcartesian leg along* $\Psi'_k(f)$ *is* $(j^k)^{-1}$ *— k-ary relation reflection,* $\{(v_1, \ldots, v_k) : (jv_1, \ldots, jv_k) \in R'\}$*. This is Lemma 8 verbatim with* $\mathrm{nodes}(T)$ *replaced by* $\mathrm{nodes}(T)^k$*, so* Ψ'_k *reproduces the arboreal k-ary fibre exactly.*

Proof. The fibre and the opcartesian leg are the §2 formulas with position set $\mathrm{nodes}(T)^k$; the proof of Lemma 8 used only that positions form a set and that inverse image is functorial. (Verified: $(j^2)^{-1}$ equals relation reflection on all sampled binary relations on the V-tree, and the Cont fibre $2^{n^2} = 16$ *is* the arboreal 2-ary fibre; `scratch/crown_stage2_repair_verify.py`.) □

So the former clause (3) "ceiling" is *false*, and we withdraw it. For a signature σ, take one shape per relation symbol r with positions $\mathrm{nodes}(T)^{\mathrm{ar}(r)}$; a coproduct over *symbols* is correct here, because distinct symbols are independent (this is a genuine coproduct, unlike the spurious branch-coproduct of §4). If the intended k-ary fibre is the Ehrenfeucht–Fraïssé one — related tuples must

be pairwise comparable, on a common root chain — take the positions to be the *comparable* k-tuples; then the earlier count 2^{n^k} is also wrong and should read $2^{\#\{\text{comparable } k\text{-tuples}\}}$. In every case the k-ary structure sits *inside* $\mathsf{Cont}(\mathrm{cod})$, at the container with positions $\mathrm{nodes}(T)^k$; what survives of clause (3) is only the trivial observation that the *unary* Ψ' cannot see it.

7 Scope, open problems, and placement

What is proved. Clause (1) — the positive one-shape extension over $\mathcal{T}$, via the *contravariant* $\Psi' \colon \mathcal{T}^{\mathrm{op}} \to \mathsf{Cont}$ with j carried backward — is *proved, unconditional* (Proposition 9), its one ingredient discharged by the six-line Lemma 8 (Rick's; no appeal to Jakl–Reggio). Clause (2)'s over-count $2^{\widehat{U}} \supsetneq 2^{\mathrm{nodes}}$ with factor $2^{\sum_v (\lambda(v)-1)}$, and its descent reading, are proved (Lemma 13, Proposition 16, Examples 14, 15), as a statement about the *object* assignment Ψ; and Ψ extends to no functor covering the subtree embeddings (Theorem 18), nor any acting canonically on automorphisms (Theorem 19). Of clause (3), only the trivial statement survives (Proposition 21); the ceiling is withdrawn (Proposition 22). The objectwise over-count has a machine-checked Lean oracle (the registry node `lean-crown-stage2-overcount-identity`).

On Jakl–Reggio. The earlier version attributed the cartesian-lift fact to T. Jakl and L. Reggio, *On the Axioms of Arboreal Categories*, arXiv:2603.21841 (CMCS 2026), Theorems 23 and 26. We do *not* use it: the fact has the direct six-line proof of Lemma 8 inside our own coloured category $\mathcal{A}$, needing no arboreal axioms (which, moreover, we never verified for $\mathcal{A}$ — our base $\mathcal{T}$ is rooted, exactly the case where the older connectedness axiom fails). The citation is corrected here and recorded as not needed.

Open problems.

1. **Proof-relevance.** The match is propositional. The full container fibre $(\mathsf{Set}/\mathrm{nodes})^{\mathrm{op}}$ is strictly richer than $\mathrm{Sub}(\mathrm{nodes})^{\mathrm{op}}$; a proof-relevant comparison would need an arboreal theory with witness-set colours.

2. **The descent completion.** Proposition 16 is elementary (a single equalizer per tree). A statement "$\mathbb{P}$ is the descent completion of $\mathsf{Cont}(\mathrm{cod}) \circ \Psi$" can *not* be phrased as a comparison of functors, since Ψ extends to no functor covering the embeddings and none natural on automorphisms (Theorems 18, 19); it must be stated as descent along the *span* R_f. That is the honest form of the open problem.

3. **Beyond embeddings.** $\mathcal{T}$ here has subtree embeddings only; open maps and non-monic tree morphisms are untreated, inherited from the Stage 1 conventions.

Placement. The net correction to the registered Stage 2 conjecture is clean. It is *not* "Ψ positive on the unary fragment, failing at $k \geq 2$". It is that the branch-multishape Ψ fails already at $k = 1$ *branching* (unfolding versus descent: no functorial action covers the embeddings, nor acts canonically on automorphisms); the positive Stage 2 holds over the full tree base via the contravariant one-shape Ψ'; and $k \geq 2$ is no obstruction at all, merely invisible to the *unary* functor. The reusable principle, worth carrying into the applied side of the programme: *when encoding a branching process in the container bifibration, do not make the branches into shapes.* That unfolds the process and over-counts shared history by one bit per extra branch through each shared node;

keep one shape and put the whole state set in the positions, because the sharing — shared provenance, shared sub-trajectory, composition itself — is a descent datum that the Fam $= \Sigma$ coproduct structurally cannot see. This is the container-side statement of "composition is a degree-two datum", and it applies verbatim to supply-chain provenance models and to agent-orchestration logs: shared sub-trajectories must be glued by an explicit descent step, not recovered from a product over branches.

References

[1] D. Ahman, J. Chapman, T. Uustalu, *Directed containers as categories*, in *Proc. MSFP*, EPTCS, 2016.

[2] N. Gambino, J. Kock, *Polynomial functors and polynomial monads*, Math. Proc. Cambridge Philos. Soc. **154** (2013), 153–192.

[3] S. Abramsky, L. Reggio, *Arboreal categories: an axiomatic theory of resources*, Log. Methods Comput. Sci. **19** (2023).

[4] T. Jakl, L. Reggio, *On the axioms of arboreal categories*, arXiv:2603.21841 (CMCS 2026). (Theorem 23: a morphism is Cartesian iff a pathwise embedding; Theorem 26: the path functor is a Street fibration. *Not used in this note*; the cartesian-lift fact is proved directly in Lemma 8.)

[5] MacBeth, *The logic of containers:* $\mathsf{Cont}(\mathrm{cod}) = \mathrm{Fam}(\mathrm{cod}^{\mathrm{op}})$ *as a bifibration*, Kodamai internal proof note, 28 Aug 2026.

[6] MacBeth, *Game comonads are polynomial comonads* (Lemma 4.1: coalgebras of a forest comonad are copresheaves), Kodamai internal proof note, 1 Oct 2026.

[7] MacBeth, *Refined Bridge-3 crown, Stage 1: the orientation gap — relation-reflection is* ρ^**, not* $(\Sigma_\rho)^{\mathrm{op}}$, Kodamai internal proof note, 2 Oct 2026.